Modular VHF-to-C-band Spectrum-Sensing Node

Baseline node. The proposed platform is a modular, single-channel scanning node for VHF-to-C-band spectrum sensing. The architecture follows the design philosophy of SOSA (Sensor Open Systems Architecture) and related open-system approaches: standard RF interfaces, replaceable front-end modules, software-defined signal processing, and clear separation between the RF front-end, timing subsystem, and digital backend.

USRP B206mini-i.

- 1×1 SDR covering 70 MHz–6 GHz, with up to 56 MHz instantaneous bandwidth and high-speed USB 3.0 streaming. The device is well suited to swept or task-driven spectrum sensing over VHF, UHF, S-band, and lower C-band regions. It operates as a half-duplex channel and should be used here in receive-only sensing mode.
- Estimated price: 1,850 USD (approximately 1,600 USD for the board-only option) •.

Jetson Orin Nano Super Developer Kit.

- The Jetson Orin Nano is selected as the embedded processing backend for FFT-based sensing, local control, metadata handling, storage coordination, and optional GPU-assisted signal classification. For fixed-site or multi-SDR deployments, an x86 mini-PC backend may still be preferable when USB stability, long-term driver maintainability, and storage capacity are more important than compact embedded-GPU operation.
- Estimated price: 250 USD.

Antenna System Configuration and RF Band Planning

The RF front-end is organized as a modular, switchable chain so that the receiver can operate either in a high-sensitivity mode or in a high-dynamic-range mode, depending on the local RF environment. In dense urban deployments, the node must avoid compression caused by strong FM broadcast, television, cellular, telemetry, or radar-like signals before those signals reach the SDR input stage. Therefore, protection and frequency-selective filtering are placed before high-gain amplification.

Antenna → Limiter/ESD protection → Switchable filter/preselector → LNA or bypass → Step attenuator → SDR

This ordering is technically preferable because the limiter and preselector reduce the probability that strong out-of-band signals will drive the LNA or SDR front-end into compression. The LNA stage must be switchable or bypassable, since a fixed high-gain path may saturate in strong-signal environments before a post-LNA attenuator can protect the receiver chain.

Band	**Antenna Type**	**Model Example**	**Frequency Range**	**Gain**
VHF/UHF survey	Biconical	Aaronia BicoLOG 5070 100 USD •	70 MHz–700 MHz	∼0 dBi
UHF–C-band survey	Log-periodic	Aaronia HyperLOG 7060 320–540 USD •	700 MHz–6 GHz	5–7 dBi

Table 1: Antenna selection for the modular multi-band spectrum-sensing node.

Supporting Hardware

i) **Low-noise amplifiers and bypass strategy.**

The LNA subsystem is divided into frequency-dependent gain paths. Each LNA path should include a bypass state so that the receiver can avoid front-end compression when strong in-band or out-of-band signals are present. This is especially important near FM broadcast transmitters, cellular base stations, telemetry sources, or other high-power emitters.

Model	Range	Gain	NF	Use Case
Mini-Circuits ZX60-33LN-S+ 159 USD •	50 MHz–3 GHz	20 dB	3 dB	VHF, UHF, and lower microwave sensing.
Additional 3–4 GHz LNA 50–150 USD	3 GHz–4 GHz	15–25 dB	TBD	Coverage gap between the two principal LNA paths.
Amplitech AGLNA035082 35 USD •	3.5 GHz–8.2 GHz	20 dB typ.	0.9 dB	Upper microwave and C-band sensing.

Table 2: Frequency-dependent LNA paths. Each gain path should include a bypass state.

ii) **Automatic antenna and RF-path switching system.**

The switching subsystem should be treated as a complete RF-path selection network, not only as an antenna selector. Depending on the final front-end layout, the design may require one switching stage for antenna selection, one stage for preselector selection, and one stage for LNA/bypass routing.

- **RF switch:** Mini-Circuits ZSWA-4-46DR+ 450 USD •.
  - SP4T absorptive switch, DC–6 GHz.
  - Insertion loss: <2.5 dB at 6 GHz.
  - Control: TTL logic, with level shifting as required by the embedded controller.
  - Function: antenna, filter, LNA, or bypass-path selection, depending on the implemented RF layout.

Note: The SP4T switch should be interpreted as the minimum switching element for the prototype. A complete modular front-end may require a cascaded switch network. Therefore, the final switch count should be defined after the RF path matrix is specified.

iii) **Preselection and band plan.**

Preselection is required to prevent overload in realistic RF environments. Rather than relying on one fixed wideband path, the front-end should use a modular filter bank. This allows the system to select the appropriate RF path for broad survey operation, protected urban monitoring, or focused analysis of a specific communications or broadcasting band.

The RF front-end should support at least three operating modes:

(a) **Wideband survey mode:** moderate gain, conservative attenuation, and broad preselection for spectrum occupancy mapping.

(b) **Protected urban mode:** FM notch enabled when appropriate, LNA bypass available, and increased attenuation to avoid compression.

Sensing Region	Suggested Filter Path	Nominal Band	Typ. IL	Purpose and Comments
70–88 MHz	Low-VHF preselector or broad VHF path	70–88 MHz	TBD	Lower-VHF sensing below the FM broadcast band.
88–108 MHz	Switchable FM notch or FM pass path	88–108 MHz	<1.5 dB pass	The FM notch protects the receiver during wideband surveys, but must be bypassed when FM broadcast monitoring is the target.
108–700 MHz	Mini-Circuits SHP-50+ + SLP-750+ ●	108–700 MHz	$\sim$2 dB	General VHF/UHF monitoring with reduced low-frequency and out-of-band loading before amplification.
700–960 MHz	Modular BPF or cavity filter	700–960 MHz	<2 dB	Mobile, public-safety, telemetry, and upper-UHF services. A replaceable SMA filter module is preferred over a fixed custom-only part.
1.7–2.7 GHz	Custom BPF or partial VBFZ-2130-S+ path ●	1.7–2.7 GHz	<2 dB	Common cellular, ISM, and microwave communication regions. The final filter should match the exact services to be monitored.
3.0–4.0 GHz	Additional BPF + dedicated LNA path	3.0–4.0 GHz	TBD	Closes the 3.0–3.5 GHz coverage gap and supports upper S-band and lower C-band sensing.
4.0–6.0 GHz	Marki Microwave C-band BPF or equivalent ●	4–6 GHz	<3 dB	Focused C-band sensing. This module should remain replaceable because C-band monitoring requirements may vary by application and region.

Table 3: Modular RF band plan for VHF-to-C-band spectrum sensing.

(c) **Focused band-analysis mode:** narrow preselector enabled, LNA path selected according to the target band, and attenuation adjusted to keep the receiver chain within its linear operating range.

iv) **Digitally controlled step attenuator.** Mini-Circuits ZX76-31+ 180 USD •.

- Attenuation range: 0–31 dB in 1 dB steps, DC–6 GHz.
- Placement: after the LNA/bypass selection stage and before the SDR input.
- Function: protects the SDR input stage and supports automatic gain management, provided that the preceding LNA and filter stages are not already compressed.
- Control: SPI through the embedded controller GPIO or through a USB-SPI bridge.

v) **Timing and synchronization.** Leo Bodnar Precision GPSDO 150 USD •.

- Outputs: 10 MHz and 1 PPS.
- Frequency accuracy: <25 ppb when GPS locked.
- Connection: USRP B206mini-i external REF IN through SMA.
- The GPS lock state should be logged as part of the sensing metadata.

Total Budget Assessment of the Bill of Materials

The estimated budget must distinguish between the core sensing node and the integration costs required to make the system operational in a real RF environment. Although the SDR, embedded processor, antennas, filters, LNAs, switching elements, attenuator, and timing reference define the main bill of materials, additional costs are required for RF cabling, connectors, enclosure, power regulation, mounting, cooling, and calibration accessories.

Key budget risks.

1. **Filter-bank uncertainty:** the FM notch filter, 700–960 MHz filter, 3–4 GHz filter path, and 4–6 GHz C-band filter may vary significantly in price depending on whether catalog or custom parts are used.

2. **RF switching complexity:** one SP4T RF switch may be sufficient for a minimal prototype, but a complete antenna/filter/LNA/bypass network may require additional switches or relays.

3. **3–4 GHz coverage gap:** the revised design closes this gap with an additional LNA and filter path. This improves technical coverage but increases the budget.

4. **Integration accessories:** RF cables, SMA adapters, terminators, enclosure, power supplies, and USB/GPIO/SPI interfaces are often omitted from early BOMs but are necessary for a deployable system.

5. **Calibration cost:** if the system is intended for repeatable or regulatory-oriented measurements, calibration accessories and verification procedures should be treated as part of the system cost, not as optional extras.

6. **Scalability cost:** each additional sensing node will require its own SDR, antenna paths, front-end protection, timing reference or timing distribution, enclosure, and calibration record.

Overall, a minimal laboratory prototype may be assembled closer to 4,700–5,000 USD if catalog parts are available and integration is kept simple. However, a more realistic and defensible planning range for a modular, field-usable VHF-to-C-band sensing node is approximately 5,400–7,200 USD, including contingency and integration overhead.

Subsystem	Estimated Cost	Comments
USRP B206mini-i SDR	1,850 USD	Main SDR receiver. Provides 70 MHz–6 GHz coverage, USB 3.0 streaming, and compatibility with an external 10 MHz reference input.
Jetson Orin Nano Super Developer Kit	250 USD	Embedded processing platform for FFT-based sensing, local control, metadata handling, and optional edge-AI classification.
Antenna system	420–640 USD	One VHF/UHF biconical antenna and one UHF–C-band log-periodic antenna. Final cost depends on vendor and availability.
Low-noise amplifier paths	244–344 USD	Includes the 50 MHz–3 GHz LNA, the upper microwave/C-band LNA, and an allowance for the 3–4 GHz LNA coverage gap.
RF switch and path selection	450 USD	SP4T absorptive RF switch for antenna, filter, LNA, or bypass-path selection. Additional switches may be required if the filter bank becomes more complex.
Preselection filter bank	595–745 USD	Includes VHF/UHF filtering, 700–960 MHz filtering, 1.7–2.7 GHz partial filtering, 3–4 GHz allowance, 4–6 GHz C-band filtering, and an estimated FM notch filter cost. Custom filters remain a budget risk.
Digitally controlled step attenuator	180 USD	Used after the LNA/bypass stage and before the SDR to support gain management and ADC-input protection.
GPSDO timing reference	150 USD	Provides 10 MHz and 1 PPS references for frequency stability, synchronization, and traceable sensing metadata.
RF integration accessories	150–300 USD	SMA cables, adapters, terminators, bias tees if required, USB 3.0 cable, GPIO/SPI wiring, and RF interconnects.
Power, enclosure, and thermal integration	200–500 USD	Regulated supplies, low-noise LDOs for RF stages, enclosure, mounting hardware, airflow, and field-deployment protection.
Calibration and test allowance	200–600 USD	Reference loads, attenuators, adapters, verification measurements, and basic calibration accessories.
Estimated subtotal	4,689–6,009 USD	Expected hardware-only range before contingency.
Contingency, 15–20%	703–1,202 USD	Recommended due to custom filters, RF accessories, shipping, taxes, price variation, and integration uncertainty.
Defensible project budget	5,400–7,200 USD	Recommended planning range for a complete, modular, field-usable VHF-to-C-band spectrum-sensing node.

Table 4: Total budget assessment for the modular SDR spectrum-sensing node.

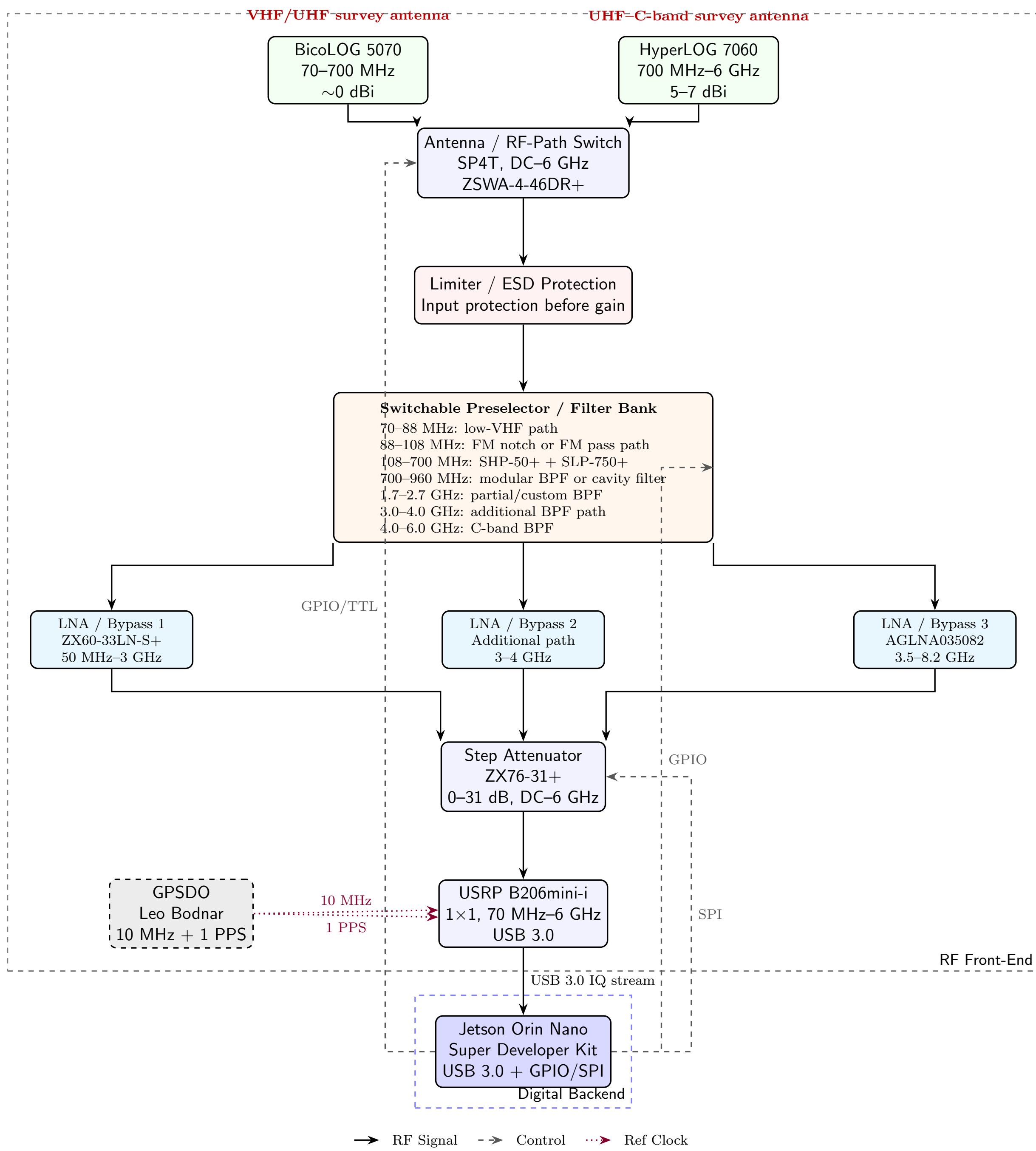

Figure 1: System block diagram of the modular VHF-to-C-band SDR spectrum-sensing platform. The architecture places limiter/ESD protection and switchable preselection before the LNA-or-bypass stage, includes the 3–4 GHz coverage path, uses the USRP B206mini-i as a 1×1 SDR, and uses the Jetson Orin Nano as the digital backend.

Power Budget Assessment

External DC input → DC/DC conversion → separate low-noise RF and digital rails

The power budget must distinguish between passive RF elements, low-power RF control components, active RF gain stages, the SDR, and the embedded processing backend. Passive antennas and passive filters do not require DC power, but they affect the RF link budget through insertion loss. Active components such as LNAs, RF switches, digitally controlled attenuators, the SDR, GPSDO, and embedded processor determine the electrical supply requirements of the sensing node.

Supply-rail architecture. The sensing node should not be powered from a single unconditioned 5 V bus. Instead, the design should separate the digital processing load from the low-noise RF front-end supply. A recommended supply organization is given in Table 6.

Operating modes. The power demand depends on the selected sensing mode.

A practical field-deployable node should therefore be designed around a **5 V supply capable of at least 8–10 A**, or an equivalent DC input architecture with local 5 V regulation. For robust deployment, a **50–60 W power subsystem** is recommended. This provides sufficient headroom for peak Jetson workload, SDR streaming, RF front-end expansion, cooling, storage activity, and startup transients.

The most important power-design constraint is not only total wattage but also supply cleanliness. The RF front-end, especially the LNA and GPSDO paths, should be powered through low-noise regulation and physically routed away from the high-current digital processing rail. This reduces the risk of supply-induced phase noise, spurious responses, gain variation, and unstable sensitivity during spectrum-sensing operation.

Subsystem	Supply Rail	Typical Power	Maximum Power	Power-Budget Notes
Biconical antenna	–	0 W	0 W	Passive antenna. No DC supply required.
Log-periodic antenna	–	0 W	0 W	Passive antenna. No DC supply required.
Passive preselection filters	–	0 W	0 W	Includes FM notch, VHF/UHF filters, cellular-band filters, and C-band filters. These elements consume no DC power but introduce insertion loss.
Mini-Circuits ZX60-33LN-S+ LNA	5 V RF rail	0.3–0.5 W	0.7 W	Low-noise supply recommended. The path should be switchable or bypassable to avoid compression in strong-signal environments.
Additional 3–4 GHz LNA path	5 V RF rail	0.3–0.8 W	1.0 W	Allowance for the added gain path required to close the 3.0–3.5 GHz coverage gap. Exact value depends on the selected LNA.
Amplitech AGLNA035082 LNA	5 V RF rail	0.3–0.7 W	1.0 W	Upper microwave and C-band LNA path. Low-noise regulation is recommended to avoid coupling supply noise into the RF chain.
RF switch network	5 V control/RF rail	0.3–0.8 W	1.0 W	Includes the SP4T RF switch and allowance for additional switching if the filter-bank and LNA-bypass network requires more than one RF switch.
Digitally controlled step attenuator	5 V or control rail	0.1–0.3 W	0.5 W	Used for gain management before the SDR input. Digital control may be provided through SPI or a USB-SPI bridge.
GPSDO timing reference	5 V rail	0.5–1.0 W	1.5 W	Provides 10 MHz and 1 PPS references. GPS lock state should be logged as part of the sensing metadata.
USRP B206mini-i SDR	5 V USB rail	3.0–4.0 W	4.5 W	Bus-powered through USB 3.0. A reliable USB 3.0 port or powered USB infrastructure is required for stable streaming.
Jetson Orin Nano Super Developer Kit	5 V system rail	7–15 W	25 W	Main embedded processing backend. Power depends strongly on CPU/GPU load, FFT processing, storage activity, and edge-AI inference workload.
Storage, USB, GPIO/SPI adapters, and cooling	5 V auxiliary rail	1.0–2.0 W	4.0 W	Allowance for NVMe or microSD activity, USB peripherals, level shifters, fan, and control interfaces.
Estimated system total	–	**13.1–25.1 W**	**40.7 W**	Estimated electrical load before design margin.
Recommended design margin, 25–30%	–	**16.5–32.5 W**	**52–55 W**	Recommended margin for startup current, temperature variation, USB peripherals, fan load, and future RF-path expansion.

Table 5: Estimated power budget for the modular VHF-to-C-band SDR spectrum-sensing node.

Rail	Recommended Rating	Loads	Notes
Main system rail	5 V, 8–10 A	Jetson Orin Nano, cooling, storage, control electronics	Provides sufficient margin for the embedded processor under GPU load and for auxiliary digital devices.
SDR USB rail	5 V, 1.5–2 A	USRP B206mini-i	Prefer a high-quality USB 3.0 connection. Avoid powering the SDR through weak hubs or long low-quality USB cables.
Low-noise RF rail	5 V, 1.5–2 A	LNAs, RF switch network, attenuator, GPSDO	Should be derived through low-noise regulation and kept separate from noisy digital loads when possible.
Control rail	3.3 V / 5 V logic	GPIO, SPI, level shifters, switch control	Used for digital control of RF switch states, attenuator settings, and metadata acquisition. Logic-level compatibility must be verified.

Table 6: Recommended supply-rail organization for the SDR sensing node.

Mode	Typical Power	Maximum Power	Description
Idle / standby monitoring	13–16 W	25 W	Jetson in low-power mode, SDR active or intermittently active, minimal RF switching, no heavy GPU inference.
Wideband survey mode	18–25 W	35 W	Continuous SDR streaming, FFT processing, moderate storage activity, and one active RF gain/filter path.
Focused analysis mode	20–30 W	40 W	Narrower preselection, active LNA path, continuous spectral estimation, event logging, and optional demodulation or classification.
Edge-AI classification mode	25–35 W	45 W	SDR streaming plus GPU-accelerated inference, local feature extraction, and higher processor load.

Table 7: Estimated power demand under representative sensing modes.

Compute pipeline

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SDR Hardware → UHD / SoapySDR → GNU Radio or Python/C++ DSP → SigMF metadata → Database / dashboard
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Use:

- UHD for USRP devices,
- SoapySDR as a hardware abstraction layer,
- GNU Radio for flowgraphs and rapid prototyping,
- Python/C++ DSP modules for reproducible sensing algorithms,
- SigMF for metadata and recorded IQ datasets,
- Docker/Podman for deployable software containers.